

Spectators of AI : Football Fans vs. the Semi-Automated Offside Technology at the 2022 FIFA World Cup

Sruthi Viswanathan
University of Oxford
Oxford, United Kingdom
sruthi.viswanathan@cs.ox.ac.uk

Jack Jackson
University of Oxford
Oxford, United Kingdom
jack.jackson@cs.ox.ac.uk

Ravi Shankar
University College London
London, United Kingdom
r.shankar@ucl.ac.uk

Reuben Binns
University of Oxford
Oxford, United Kingdom
reuben.binns@cs.ox.ac.uk

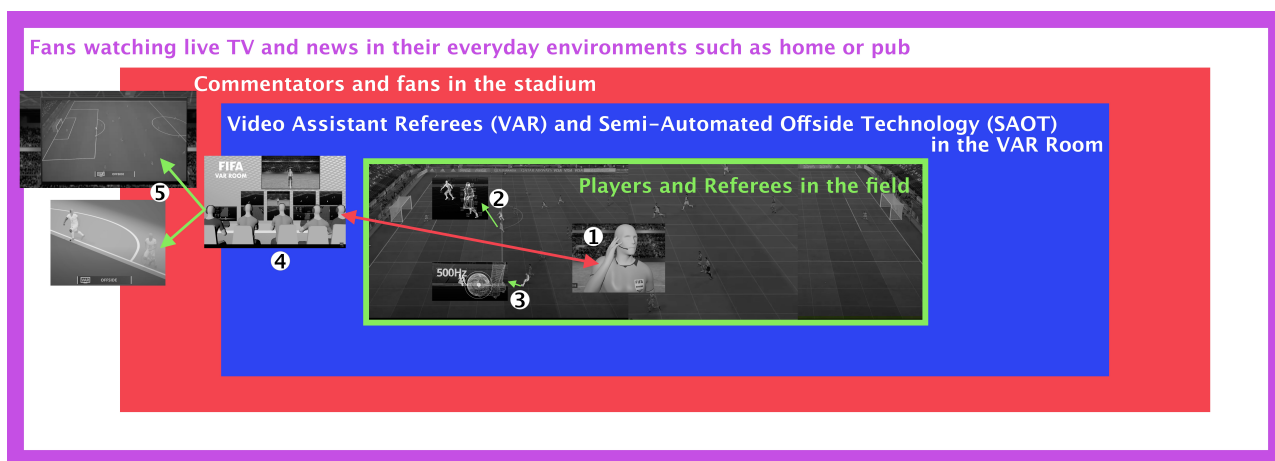


Figure 1: Illustration of events succeeding an offside call at the FIFA World Cup 2022, with the on-field referee ① making a call to the Video Assistant Referees (VAR) ④ to check for offside. The VAR room then use the Semi-Automated Offside Technology (SAOT), which receives data from 12 cameras tracking the players ② and from the sensor inside the match ball ③. Considering the data from SAOT, a decision is reached by the on-field referee ① and VAR room ④. This decision is then communicated to the sports commentators and fans using a 3D animation ⑤.

ABSTRACT

More than half of the global population were spectators of AI-assisted decision making during the recent 2022 FIFA World Cup. For the first time in a Football World Cup, *via* news and social media, fans were briefed about the Semi-Automated Offside Technology (SAOT), which determined the position of players to be offside or not, in real time, with the help of AI. In this paper, we collect and analyse news articles related to offside decisions made with SAOT, published during and around the tournament. Our findings reveal alternating positive and negative portrayals of SAOT in the news, gaps in communication, and needs of Football fans to gain a better spectator experience. In conclusion, we discuss general

implications for designing a better relationship between Spectators and AI, specific implications for bettering the experience of Football fans when using SAOT, and introduce our future research.

CCS CONCEPTS

• **Human-centered computing** → **Displays and imagers**; • **Information systems** → **Decision support systems**.

KEYWORDS

spectators of AI, interactive AI, explainable AI, spectator sports, Football, soccer, offside

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1 INTRODUCTION AND RELATED WORK

On 18 December 2022, Argentina won the *Federation Internationale de Football Association* (FIFA) World Cup 2022. There were multiple turning points during this nerve-racking Football (a.k.a Soccer) match which contributed to Argentina's victory, one among them was a critical offside decision, made with the help of FIFA's Semi-Automated Offside Technology (SAOT [6]). With the help of SAOT, referees had allowed Argentina's third goal in the 108th minute. Similar to most offside decisions made throughout this World Cup tournament, this decision was labelled to be *controversial* by billions of Football fans, who were active spectators of AI-assisted decision making during this match. Fans took to various social media platforms to voice their outrage or help clarify the decision to fellow fans.

Offside is one of the codified laws [11] in Football, that are maintained and updated by International Football Association Board (IFAB), used to prevent a player from gaining an unfair advantage by being in a position closer to the opponent's goal than the ball and the second-to-last defender (usually the last defender is the goalkeeper), when their team-mate plays the ball forward to them. If the player scoring a goal is found offside, then the goal is consequently disallowed. SAOT was designed to assist the Video Assistant Referees (VAR) and on-field referees in making offside decisions (See Figure 1). SAOT was introduced by FIFA as the AI technology that will make faster, more accurate, and more reproducible offside decisions [6]. SAOT utilizes a system of 12 cameras placed under the stadium's roof to track players, and a motion sensor, called Inertial Measurement Unit (IMU), located in the center of the official World Cup ball. These cameras use up to 29 data points per player, primarily focusing on the face and extremities of the limbs, and are monitored 50 times per second to accurately determine the players' positions on the pitch at all times. Additionally, the IMU sends data up to 500 times per second, allowing for precise detection of the kick point, without being noticeable to players. These technologies allow match officials to review live data for the first time, automatically providing very accurate information on where a player is, and when a player has touched the ball, ultimately improving the quality and speed of VAR decision making in regards to offside situations and unclear touches.

SAOT was announced by FIFA several months before the World Cup, who also made subsequent press releases to dissuade journalists from terming SAOT as *Robot offside* [5]. FIFA highlighted on the fact that the final decision was indeed made by human referees. Based on the feedback from their "fan experience panel," user research with 25 participants, a 3D animation showcasing the offside situation was to be displayed for both in-stadium fans and television viewers [4]. While this was the original plan, within a few minutes into the first World Cup match, SAOT was called biased (See A5 in Table 1). Football fans had to largely rely on the commentators to understand the offside decisions made with the help of SAOT, the precision of which was invisible to their human eyes. As a result, around 5.4 billion people who tuned in to watch the FIFA World Cup from all around the world, took away a negative mental model of AI, as a complicated and controversial decision maker, an extended version of our biased society.

Lack of transparency negatively affects end users' sentiments about an AI-assisted future and also poses a potential threat to fair play in sports. Although application of AI in sports industry is less critical compared to domains such as lawmaking and public health, where the general public is heavily impacted by the decisions made, it is still important to strive for a better and enhanced spectator experience with AI in kinder domains to set and manage societal expectations of AI. The use of AI in sports [7, 10, 16] and AI in Football [9, 14, 17] have been discussed in the literature. The potential of AI for Football [14, 17] and various other sports [7], advancements of specific techniques such as bio mechanics [16], applications in particular tasks such as refereeing [9], a review of the current advances [16], and accounts of sports scientist on the current challenges for AI advancement in sports [10], are the leading topics of prior research on AI in Sports. Paradoxical tensions between various stakeholders, including fans, on the topics of accuracy, objectivity, explainability, relationship with artistry, on using AI in sports such as gymnastics [15] has also been debated in the literature.

Our research goal was to understand spectators' sentiments and needs on being subjected to AI-assisted decision making during Football games, specifically for offside decisions, and derive implications for improving the experience for spectators of AI. While scrapping fans' social media posts would have given us a first-hand account of their perception of SAOT, we chose to analyse news articles, as journalist prepare their work for scrutiny and analysis [13], whereas social media users do not want their posts used for research without their consent [8]. Previously, social perception of AlphaGo, a computer program that plays the game of Go (the most challenging classical game for artificial intelligence), had been analysed in the literature via news articles [3, 13].

Therefore, we analysed the portrayal and reception of SAOT during the FIFA World Cup 2022 by analysing 366 news articles from around the world, our findings reveal that articles published prior to the World Cup were highly positive with explanations about the helpfulness of AI, articles post World Cup to be mostly neutral with debriefing analyses of decisions made, and a contrasting set of articles published during the course of the World Cup to be highly negative discussing the outrage of spectators and other stakeholders (such as players and commentators) on the offside decisions. Our findings also highlighted the confusions caused due to the communication gaps between the offside decision makers and spectators and the needs of spectators with regards to this situation. Based on these findings, we discuss our design implications to improve the relationship between spectators and AI. Specifically, we also focus our discussion on creating a better experience for Football fans when using SAOT, to ensure a clear and consistent communication that make offside decisions more transparent to fans, and to engage them by soliciting their feedback. We also discuss our future research plans to investigate further into spectators' biases on AI-assisted decisions and design for a better presentation of information pertaining offside calls.

2 ANALYSIS OF NEWS ARTICLES

Using a Python script built with the `urllib.request` library, we scrapped around 1000 Google News articles in the first round, using the keywords “AI assisted offside decisions in FIFA World Cup” and “Decisiones de fuera de juego en la Copa del Mundo FIFA 2022.” After cleaning these articles to remove duplicates, instances from previous World Cups, and paraphrased articles of the same instance, we resulted with around 200 articles. We further scrapped another 1000 Google News articles using the keywords “SAOT Semi-Automated Offside Technology used in FIFA World Cup 2022,” “Controversial offside decisions in FIFA World Cup 2022,” and “Controversias de fuera de juego en la Copa del Mundo FIFA 2022.” After removing duplicates and cleaning those articles which did not discuss SAOT and/or Football fans, we were left with a total of 366 distinct articles (attached as supplementary material of this paper).

While there were thousands of articles published across the world about this billion-spectator tournament, we decided to limit our dataset to 366 articles that focused on SAOT and Football fans, to enable us to perform a qualitative analysis on them. Our dataset (See Table 1) considered of articles from 46 countries who both participated in the World Cup (such as Argentina and Morocco) or had a huge Football-fan base (such as India and Kenya). Also, as most of the articles were published during the World Cup and as we wanted to focus on the spectator experience, we sampled articles across this time period as: 20% before World Cup, 75% over the course of the 64 matches played during the World Cup, and 5% post World Cup. Two authors open coded these articles in separation for reflexive thematic analysis [1]. We resulted with a total of 63 codes which were then grouped in two rounds of meetings by these primary coders into three key themes, which were then discussed between all authors, and a consensus was reached after two meetings.

3 FINDINGS

In this section, we present the results from our analysis of 366 news articles discussing the use of SAOT in the 2022 FIFA World Cup. Our findings are organized into three main themes: (1) the variation in media portrayal of SAOT before, during, and after the tournament; (2) gaps in communication regarding the actual usage, benefits, and implementation of SAOT; and (3) the expectations and needs of spectators regarding SAOT-based offside decisions.

3.1 Positive in theory and negative in practice

In the months leading up to the World Cup, news articles generally had a positive outlook and praised the introduction of SAOT for the World Cup tournament. Most of these articles (See A1, A2, A3, and A4 in Table 1) paraphrased FIFA’s original account of SAOT [6]. The goal of these articles was to introduce the working of SAOT to spectators, along with the promises it brought such as speed, accuracy, and mitigation of human biases in offside decisions.

However, as soon as the tournament began, the sentiment of the news articles turned negative. Controversies were the primary discussion of news articles with every decision made by SAOT (See A5, A6, A7, and A8 in Table 1). In contrast with its first portrayal as a time saver (See A2 and A3 in Table 1), in actual practice SAOT

was heavily criticised for causing prolonged delays (See A6 and A8 in Table 1).

After the World Cup, news articles once again became positive or neutral, focusing on the factual analysis of the decisions (See A10 in Table 1) rather than blaming the decisions made.

3.2 Communication gaps and resulting confusions

The hope of fixing human error and biases in sport was one of the main reasons for the implementation of SAOT in the World Cup (See A4 in Table 1). However, as soon as the tournament started, SAOT was called out to be biased (See A5 in Table 1), as the reasoning behind the offside decisions could not be comprehended by spectators, who resorted to social media platforms such as Reddit and Twitter to voice their outrage (See A5 and A6 in Table 1). In an effort to clarify these decisions, explanations were circulated along with 3D images (see A5 in Table 1). Despite this, the process in which offside decisions were made appeared concealed to fans, leading to outbursts of frustration and anger. Initial explanations given about SAOT quickly became outdated and out of context in comparison with what went on in the field, and as discussed in Section 3.1 the media mainly focused on the controversial offside calls.

During the beginning of the tournament, both media and fans began to use terms such as “robot linesman” and “robot referee” to describe the technology being used (See A9 in Table 1), despite FIFA dissuading the usage of such terms [5]. This led to a biased way of looking at the technology, as it suggested that decisions were being made by machines rather than human referees, while in reality human referees had made the final decisions. In contrast, as the tournament progressed, the trial of *explaining the AI* was abandoned and news articles resorted to term “VAR offside,” indicating the Video Assistant Referees (VAR) setup, which was the predecessor of SAOT (See A7, A9 and A10 in Table 1). Thus, further confusing spectators about the decision-making process and not reporting on the referee-SAOT collaboration.

Among the excessive sources of confusion, a significant and repetitive source was the use of 3D animations that misrepresented the decision to have been on finding a players’ hand offside (See Figure 2); as according to the offside laws [11] the hands and arms of all players are not considered. Despite the fact that referees had manually checked these calls, many fans were overwhelmed by the complexity of the technology and the lack of clear explanations. It is also to be noted that the frame(s) of the 3D animation to be released was chosen by the match officials.

The tournament also highlighted the dependency of fans on commentators to understand the decisions being made. However, commentators often voiced their own doubts and disagreements with the calls, adding to the confusion and frustration of fans.

3.3 Spectator needs

Spectators watching AI-assisted decision making in Football had voiced their specific needs in order to fully understand and engage with the technology, via social media platforms online. According to news articles summarising the outrage of spectators, their main

Table 1: Sample data from our 366 new-articles dataset presented in the supplementary material of this paper

Article number	Source	Title
A1	Forbes USA	Artificial Intelligence Is Changing Soccer And Could Decide The 2022 World Cup
A2	Lifestyle Asia	A closer look at the new technology being implemented for the Qatar World Cup 2022
A3	Mirror UK	FIFA bringing in new technology to make offside decisions in just three seconds
A4	Al Jazeera Doha	World Cup in Qatar to use semi-automated offside system
A5	La Nacion Argentina	Mundial 2022: cómo funciona el nuevo VAR que ahogó el grito de gol de Ecuador a los 3 minutos
A6	Marca Spain	'Polémico fuera de lugar que le invalida un penalti a Croacia
A7	Sports Breif Kenya	Footie Fans Convinced Ghana Was Robbed After Photo Showing Portugal's Offside Goal Emerges
A8	infobae Mexico	Las razones detrás de la llamativa extensión del tiempo agregado en los primeros partidos del Mundial Qatar 2022
A9	ESPN UK	Robot VAR offside disallows first goal at FIFA Club World Cup
A10	ESPN India	2022 World Cup VAR review: Every decision in Qatar analysed

**Figure 2: Controversial offside 3D animation, which appears to wrongly flag a situation as offside where the offense player (in the blue and white jersey) has their hand forward. Image credit: NPO 1.**

demand is for detailed explanations regarding multiple factors involved in the decisions being made (See A5, A6, A7, and A8 in Table 1).

Fans wanted to understand why certain instances were being reviewed, while other similar instances were ignored. They wanted to know the reasoning behind the decision, and they were even willing to learn using similar offside situations retrieved from other games as examples. They also wanted to know who are and what is involved in the decision, to help them understand that it was not biased. Some fans also questioned if the decision making process was changed as the tournament progressed.

Additionally, fans wanted a place to check the evidence, to be able to trust the fairness of the sport, and if the decisions were as reproducible as promised to them. They wanted to be able to review the same footage that was available to the referees, and to understand the steps in the decision-making process.

4 DISCUSSION

In this section, we first generalise our design discussions for spectators of AI, then focus specifically on the designing for the offside

case in Football. We also introduce our future directions and discuss our empirical experiments.

4.1 Designing for Spectators of AI

When adopting AI assistance in domains such as sports, entertainment, art, and policy making; where millions and billions of everyday people are engaged as spectators; the entire control rests with the technology creators and domain experts. As a result, the spectators' role has evolved into that of passive observation, with little opportunity for direct interaction with the technology as the events unfold.

This lack of interaction can make it difficult for spectators to fully understand the role of the technology and the changes brought by it. Also, making it difficult for them to exercise their right to accept or contest the decisions being made. Therefore, in addition to releasing prior explanations of the setup, it's important for creators and enablers of AI technologies to effectively convey the setup and process in a comprehensible and accessible format, as and when the events unfold (within the situated context), for the sake of AI spectators as much as they do for the sake of AI users.

To facilitate this understanding, a central hub or source of information could be established to gather spectators both pre- and post- event. This could be included in a typical website or app established by the event managers, where spectators can gain access to notes, video replays, images, and other evidence related to the use of AI and its impact on the event. This will help to provide context and explanations for fans, particularly when it comes to outliers or controversial incidents. This could serve as a public repository, where spectators could also be given the opportunity to register their feedback on particular instances. In order to keep spectators informed, updates on any changes made to the technology or rules should be made available.

4.2 Designing for AI adoption in Football

In January 2023, in consideration of the frustration of Football fans, IFAB has announced to trial referees providing voice overs, to explain offside decisions during the live match, in the upcoming FIFA Club World Cup 2023 [12]. While this is a step in the right direction, this puts referees in a stressful spotlight, might need

post-match amendments to offside statements and furnishing of visual evidences, which might or might not reach fans.

Based on the findings resulting from our analysis, in this section we discuss 10 implications for designing a better offside experience for Football fans.

- (1) **Naming** As the name Semi-Automated Offside Technology (SAOT) did not bind with the spectators, leading both journalists and spectators to use alternative terms such as robot referee or VAR, opting a spectator-friendly name that helps to re-imagine SAOT as a tool used by referees to check offside (such as *offside checker*), will help gain a wider adherence in the real world.
- (2) **Explanations** To help spectators to fully understand offside decisions made with the help of technology in Football, it is important for them to have access to straightforward explanations of how the calls being made (such as in the official websites of FIFA or IFAB). This could include a demo of active participants in the decision making, set in various simulated controversial offside situations, to help them better understand the steps and reasoning behind these decisions. This could also serve as the hub to update fans of the changes made to the offside tech or process.
- (3) **Intent to review** During the course of the match, succinct signals either gestured by the on-field referee, voiced by the commentators, or visually shown *via* stadium/TV screens could establish why the call to check offside is being made or skipped.
- (4) **Decision making** While the processing time for SAOT is around 20 seconds, the offside decision making process (See Figure 1) takes several minutes. These minutes, where the play is paused and fans watch replays, could be effectively redesigned to show the progress and process involving the on-field referee(s), the VAR referee(s), and SAOT. This could include providing information on the different stages of the decision-making process and the factors that were taken into account.
- (5) **Decision** When a decision has been made, the organisers only have a few seconds to communicate it during the live match. Effectively displaying the transparent reasoning during these few seconds will help mitigate rumours about awarding a biased decision. This would include considering ways to convey SOAT's suggestions and the final decisions of the referees, as well as showcasing the affordances referees have for forming a more subjective decisions based on their direct observations.
- (6) **Visualisation** Selecting an appropriate frame to visualise as the 3D animation to be displayed to the spectators is a critical for communication during the live match. This visualisation could also be augmented by labelling the body part of the player contributing to the offside, to avoid confusing the fans with hands and arms rule.
- (7) **Communication templates** Establishing a set template or guidelines for communication from the live commentators to handle an offside situation, would manage expectations, and help fans to get acquainted with the process. As the press plays an important role as a medium between organisations

and the general public, especially in the sports industry, supplying them information in a well designed offside communication template would assist in their reporting and amplify the reach of accurate knowledge to the spectators.

- (8) **Repository** Post incident, in both live communication and press releases of the offside, spectators could be given the option to retrieve the instance from an online repository maintained by the organisers (such as in the official website of FIFA or IFAB). This will increase the trust of spectators as well as entail a fair process, with fans having option to access and review the results of these decisions. Fans could be redirected by maintaining an easy to access repository and also by advertising the same with the use of QR codes in the live match and press replays, and via web links in online news articles.
- (9) **Feedback** Alongside the repository, a mechanism for online feedback accumulation could be implemented to allow fans to provide their own input and opinions on the decisions. This would welcome the fans to be the inside the loop, and would also give way for an ethical methodology to solicit spectator feedback rather than resorting to collection of unintended feedback from social media or second-handed feedback from news articles.
- (10) **Inclusion** At present, Football has more number of spectators (around 5.4 billion) than people who have access to the internet (around 4.9 billion). Beyond live match and news telecasts, to be inclusive of fans without internet access, explanations and feedback opportunities could be circulated *via* offline channels (such as newspapers and other print media), localised Football organisations, and by using phone call with Interactive Voice Response (IVR) to grant access to websites hosting explanations. An inclusive setup by the organisers (such as FIFA and IFAB), could aid in disseminating information to reach all spectators with the help of their respective local bodies.

4.3 Future Work

One dimension for researchers and practitioners to explore in future could be a deeper examination of the impact of AI on decision making in Football. Examining the current mental models of Football fans and the biases they sustain, we plan to conduct a study where participants (who are Football fans) are presented with a set of offside decisions, with half being made using AI and half being made using non-AI means. Participants would be asked to label 20 offside decisions blindly with their feedback, without knowing whether they were made using AI or not.

After labeling, Group A would be informed that decisions 1-10 were made using AI and provided with information on the specific AI decision. They would be asked to reflect on their original labeling and consider if they would have made a different decision knowing that it was made using AI. Whereas, Group B would be informed that decisions 11-20 were made using AI, but in a deceptive manner (i.e. these decisions were in truth not made by AI). They would then also be asked to reflect on their original labeling and consider if they would have made a different decision on knowing that it was made using AI. Participants' decision-making process would also

be explored through a follow-up interview and the truth about AI and non-AI examples in our sample will be disclosed at the end of the study. This approach would allow for a comparison of offside decisions agreed upon blindly with those made using AI vs Non-AI means. Additionally, it would provide insight into whether or not knowing that a decision was made using AI leads to different opinions and different reasons for disagreeing with the decision.

Another dimension for stakeholders and information designers could include a deeper examination of the design of a static spectator screen (to be used in live TV, replays, and media) for offside decisions made with SAOT. Reflecting on the 10 implications listed in Section 4.2, we plan to design a static screen that displays key information related to the decision, which could be used during the live match, in replays, and in the press. This screen could then be tested with Football fans to gather feedback on its effectiveness in communicating the decision and the reasoning behind it.

With 1:1 semi-structured interviews we could capture the personas of participants as Football fans, their self-reported level of fandom, and their biases with regards to the teams and players they support. As the next section of the interview, they would be shown offside decisions from the World Cup and asked about their level of agreement with the decision, whether they felt it was fair or unfair, and who made the decision. Participants would also be asked about any changes they would suggest to improve the communication and understanding of the technology. As experience prototyping [2], the interviews would include a section on eliciting feedback on our design of the static spectator screen, and end with a discussion on their views on the future of AI-assisted decision making in Football. These interview would provide valuable insights into the needs and perspectives of Football fans and inform future designs to improve the communication and engagement of fans with decision-making technology.

5 CONCLUSION

In this paper we examine and discuss the case of Spectators of AI, by exemplifying the recent use of AI-assisted decision making in FIFA World Cup 2022. Our research aimed to understand the impact of technology on the experience of Football fans by analysing news articles related to offside decisions made with the Semi-Automated Offside Technology (SAOT). Our findings suggest: (i) the positive introduction and explanations of the technology before the tournament and the controversial reception of it during the tournament, (ii) gaps in communication causing confusions between the organisers, match commentators, journalists, and Football fans, and (iii) the need for clear explanations, visual aids, and opportunities for fan engagement, and feedback. Deriving implications from these findings, we discuss designing a better relationship between Spectators and AI. In particular, we also discuss designing better experience for Football fans with SAOT, reinforcing the need for clear and consistent communication in order for the technology to be transparent and accepted by fans. We aim to further explore spectators' biases in regards to AI use and design visualisation of information for offside communications in the future. By addressing these issues, we believe that AI can be leveraged in a way that enhances the overall experience of spectators, rather than detracting from it.

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